Year 12 Maths - Using graphs

Similar questions - Transformations of graphs

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) The curve $y = \dfrac{1}{x}$ is translated by $\binom{-2}{3}$. Write down the equation of the new curve and the equations of its asymptotes. **[3]**

(b) The point $A(4, \frac{1}{2})$ lies on the curve $y = \dfrac{2}{x}$. This curve is stretched parallel to the x-axis with scale factor $\frac{1}{3}$. Find the equation of the new curve and the coordinates of the image of A. **[3]**

2 $\mathrm{f}(x) = x^2 - 4x + 1$.

(a) Write $\mathrm{f}(x)$ in the form $(x-p)^2 + q$ and hence state the coordinates of the minimum point of $y = \mathrm{f}(x)$. **[3]**

(b) The graph of $y = \mathrm{f}(x)$ is reflected in the x-axis and then translated by $\binom{0}{5}$. Find the equation of the resulting curve and the coordinates of its maximum point. **[3]**

(c) The graph of $y = \mathrm{f}(x)$ is stretched parallel to the y-axis with scale factor 3, and then translated by $\binom{0}{-1}$. Write down the equation of the new curve, and the coordinates of its minimum point. **[3]**

Question 1

(a) Translation by $\binom{-2}{3}$ replaces x with $x+2$ and adds 3: $y = \dfrac{1}{x+2} + 3$. Asymptotes $x = -2$ and $y = 3$

(b) $y = \dfrac{2}{x}$ becomes $y = \dfrac{2}{3x}$ (replace x by $3x$). $A \mapsto \left(\frac{4}{3}, \frac{1}{2}\right)$

Question 2

(a) $(x-2)^2 - 3$. Minimum point $(2, -3)$

(b) Reflection gives $y = -\mathrm{f}(x)$; translation gives $y = -\mathrm{f}(x) + 5 = -x^2 + 4x + 4$, with maximum point $(2, 8)$

(c) $y = 3\mathrm{f}(x) - 1 = 3x^2 - 12x + 2$, with minimum point $(2, -10)$